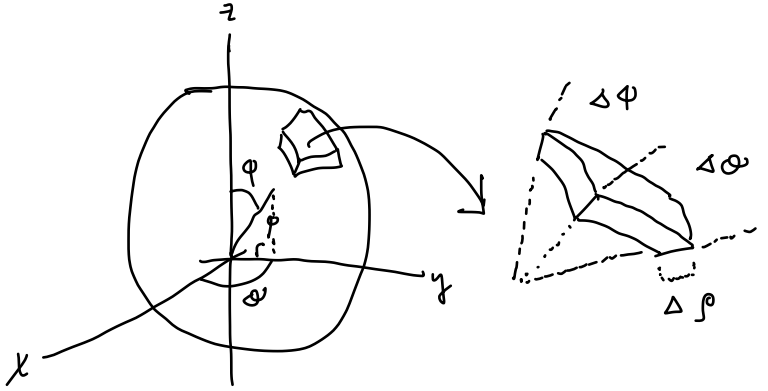


$$\Delta V_{ijk} \approx \sum_{k=1}^{l} \sum_{j=1}^{m} \sum_{i=1}^{n} \rho_i^2 \sin(\phi) \Delta\rho \Delta\phi \Delta\theta$$

$$V = \lim_{n,m,l \to \infty} \left(\sum_{k=1}^{l} \sum_{j=1}^{m} \sum_{i=1}^{n} \rho_i^2 \sin(\phi) \Delta\rho \Delta\phi \Delta\theta \right)$$

$$V = \int_0^{2\pi} \int_0^{\pi} \int_0^{R} \rho^2 \sin(\phi) \, d\rho \, d\phi \, d\theta$$

$$V = \int_0^{2\pi} \int_0^{\pi} \frac{R^3}{3} \sin(\phi) \, d\phi \, d\theta$$

$$V = \frac{R^3}{3} \int_0^{2\pi} 2 \, d\theta$$

$$V = \frac{4}{3} \pi R^3$$